Global Urban AQI Data Science Project

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- **GitHub Repository Link:** [DS Project Repository](https://github.com/Zainub042/DS)

1. Introduction

Air pollution is a critical global issue, and the Air Quality Index (AQI) provides a standardized measure of air quality. This project applies data science methods to analyze AQI data, visualize pollution patterns, and build predictive models.

2. Dataset Description

- **Dataset:**

  Global Urban AQI Dataset

- **Features:**

  AQI, PM2.5, PM10, CO, NO2, O3, SO2, Temperature, Humidity, Wind Speed, Date, City, Country.

- **Period:**

  2015–2025

- **Size:**

  `3660 rows × 13 columns`

3. Data Cleaning Steps

- Converted `Date` to datetime and extracted `Year`.
- Handled missing values using median imputation.

- Removed duplicates and invalid AQI entries.
- Scaled features using `StandardScaler`.

4. Basic Statistics

- **Mean AQI:** 164.64
- **Minimum AQI:** 30
- **Maximum AQI:** 300
- **Standard Deviation of AQI:** 78.57
- **Highest AQI Country:** India (avg AQI = 170.6)
- **Lowest AQI Country:** Australia (avg AQI = 159.6)

5. Exploratory Data Analysis (Charts)

- **AQI Category Distribution:**

  Most records fall into Moderate and Unhealthy.

- **Average AQI by Country:**

  India, Japan, Brazil highest; Australia lowest.

- **AQI Trend by Year:**

  Fluctuations across 2015–2025, often above Unhealthy.

- **PM2.5 vs AQI Scatter:**

  Strong positive correlation; PM2.5 drives AQI.

- **Correlation Heatmap:**

  Weak correlations overall, suggesting complex interactions.

6. KNN Results

- k=3 → Accuracy ≈ 60.11%
- k=5 → Accuracy ≈ 61.89%
- k=7 → Accuracy ≈ 64.07% (best)
- **Interpretation:**

  Larger k smooths decision boundaries, reducing overfitting.

7. Naive Bayes Results

- Accuracy ≈ 99.45%
- Confusion matrix shows better handling of minority classes.
- **Interpretation:**

  Outperformed KNN due to probabilistic approach.

8. K-Means Clustering Results

- 3 clusters formed, all in Unhealthy range.
- Differences lie in pollutant composition:
  - Cluster 0 → CO-dominated
  - Cluster 1 → High particulate matter
  - Cluster 2 → Mixed pollutants

9. PCA Visualization

- PC1 (10.9%) → pollution intensity (AQI, PM2.5, PM10, NO_2).
- PC2 (10.6%) → ozone + weather factors.
- First two PCs explain ≈ 21.5% variance.
- AQI categories and clusters form distinguishable regions.

10. Final Comparison Table

Method	Type	Purpose	Main Result
KNN (k=3)	Supervised	Predict AQI Category	60.11% Acc
KNN (k=5)	Supervised	Predict AQI Category	61.89% Acc
KNN (k=7)	Supervised	Predict AQI Category	64.07% Acc
Naive Bayes	Supervised	Predict AQI Category	99.45% Acc
K-Means	Unsupervised	Group similar records	3 clusters
PCA	Dimensionality Red.	Visualize data in 2D	21.5% Var

11. Conclusion

This project demonstrated that AQI is heavily driven by PM2.5 concentrations, with global averages often exceeding unhealthy thresholds. Visualizations revealed country differences, yearly fluctuations, and pollutant relationships. Among supervised models, Naive Bayes outperformed KNN, achieving nearly 80% accuracy. K-Means clustering grouped cities into three polluted profiles, while PCA provided useful 2D visualization despite limited variance explained. Overall, the analysis confirms that air quality remains a pressing issue, and machine learning can effectively classify and cluster pollution patterns.

12. GitHub Repository Link

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13. References

- Dataset source: [Global Urban Air Quality Index Dataset (2015–2025)](https://www.kaggle.com/datasets/syedmtalhahasan/global-urban-air-quality-index-dataset-2015-2025?resource=download)
- Libraries Used: pandas, numpy, matplotlib, seaborn, warnings, scikit-learn (model_selection, preprocessing, impute, neighbors, naive_bayes, cluster, decomposition, metrics)